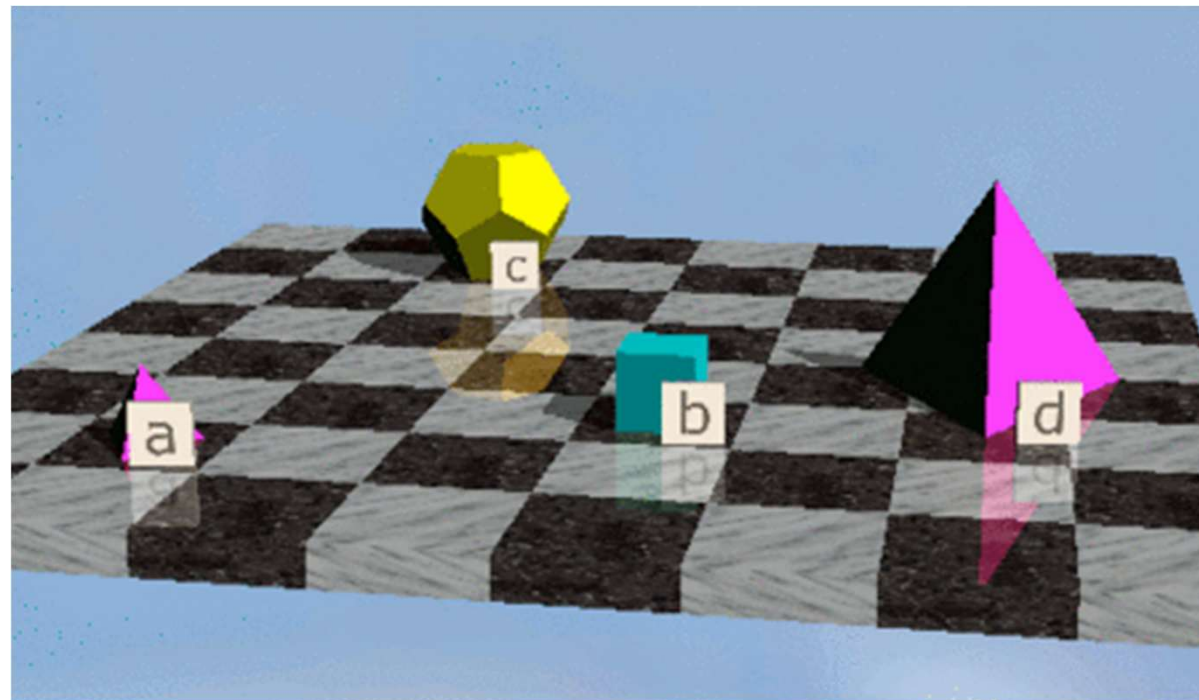


First-Order Logic



Slides from Stuart Russell, Stanford & UC Berkeley

Limited “Expressive power” of Propositional logic

- Rules of chess:

- 100,000 pages in propositional logic
- 1 page in first-order logic

- Rules of Wumpus:

- cannot say "pits cause breezes in adjacent squares"
 - except by writing one sentence for each square

- Rules of Pacman:

- $\forall t \text{ Alive}(t) \Leftrightarrow$
 $[\text{Alive}(t-1) \wedge \neg \exists g, x, y [\text{Ghost}(g) \wedge \text{At}(\text{Pacman}, x, y, t-1) \wedge \text{At}(g, x, y, t-1)]]$

First-order logic

- First-order logic (like natural language) assumes the world contains
 - **Objects:** people, houses, numbers, colors, baseball games, wars, ...
 - **Relations / Predicates:**
 - **Unary** relation or property: red, round, prime, ...
 - **n-ary:** brother of, bigger than, part of, comes between, ...
 - **Functions:** father of, best friend, one more than, plus, ...
- Some basic elements of natural language (included also in FOL):
 - Nouns and noun phrases referring to **objects**
 - Verbs and verb phrases referring to **relation** among objects
 - Some of them are **functions**

Syntax of FOL: Basic elements

- Logical elements
 - Connectives $\neg, \Rightarrow, \wedge, \vee, \Leftrightarrow$
 - Quantifiers $\forall, \exists$
- Domain specific elements
 - Constants: KingJohn, 2, NUS,...
 - Predicates Brother, >,...
 - Functions Sqrt, LeftLegOf,...
- Non-logical general elements
 - Variables $x, y, a, b, \dots$
 - Equality $=$

Possible worlds

- A possible world of five objects:

-  “left leg” unary function (arity is # arguments)

-  “on head” binary relation

-  “brother” binary relation

-  “person” unary relation

-  “king” unary relation

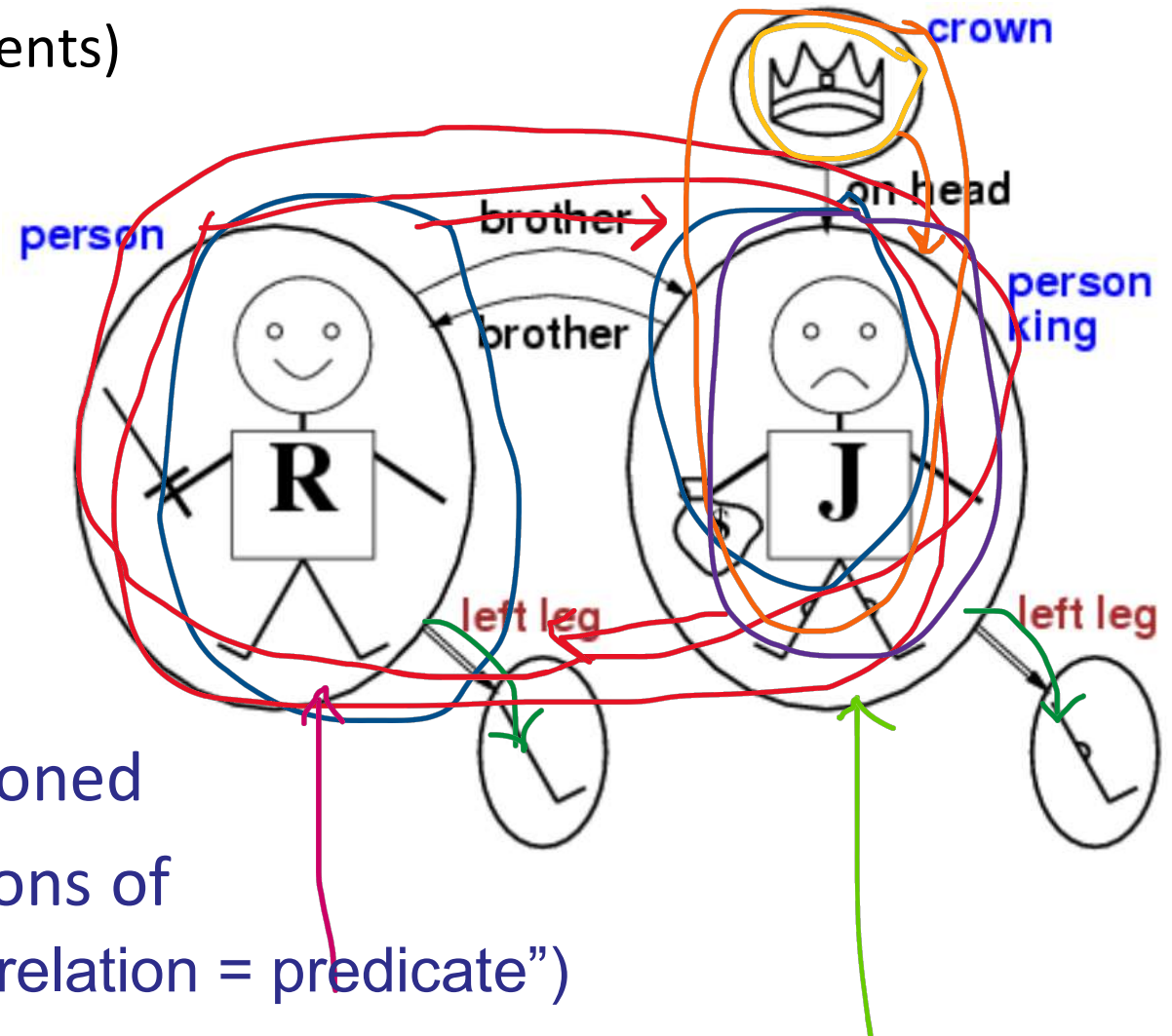
-  “crown” unary relation

-  “John” constant (0-ary function)

-  “Richard” constant (0-ary function)

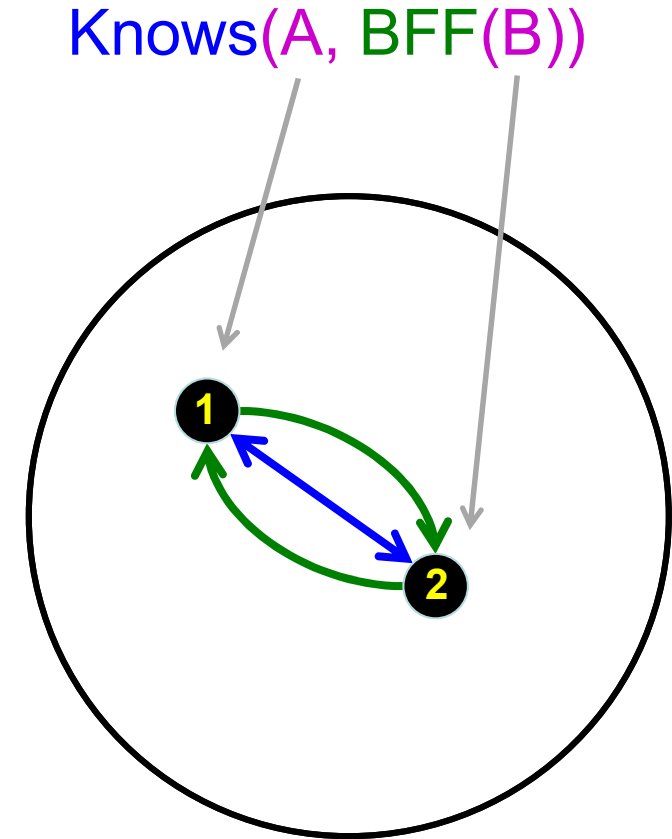
- If a function/relation/constant is mentioned

- World must have object(s) plus definitions of those functions/relations/constants (“relation = predicate”)



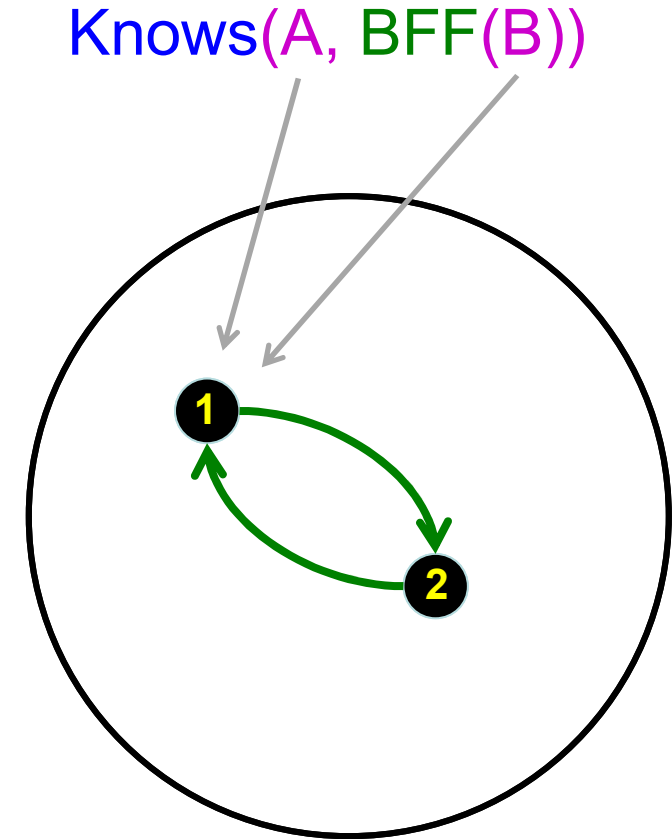
Possible worlds

- A possible world for FOL consists of:
 - A non-empty set of objects
 - For each k-ary predicate in the language, a set of k-tuples of objects (i.e., the set of tuples of objects that satisfy the predicate in this world)
 - For each k-ary function in the language, a mapping from k-tuples of objects to objects
 - For each constant symbol, a particular object



Possible worlds

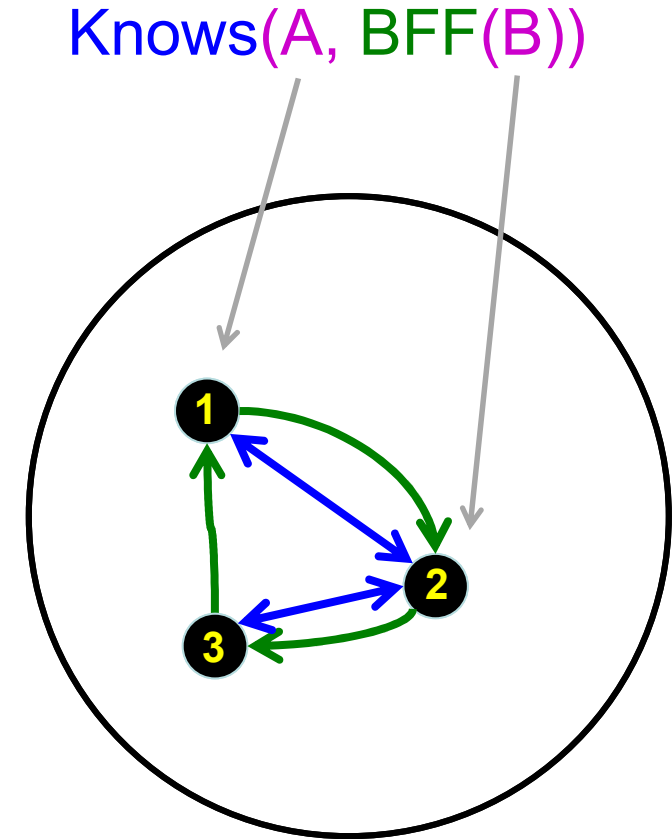
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Possible worlds

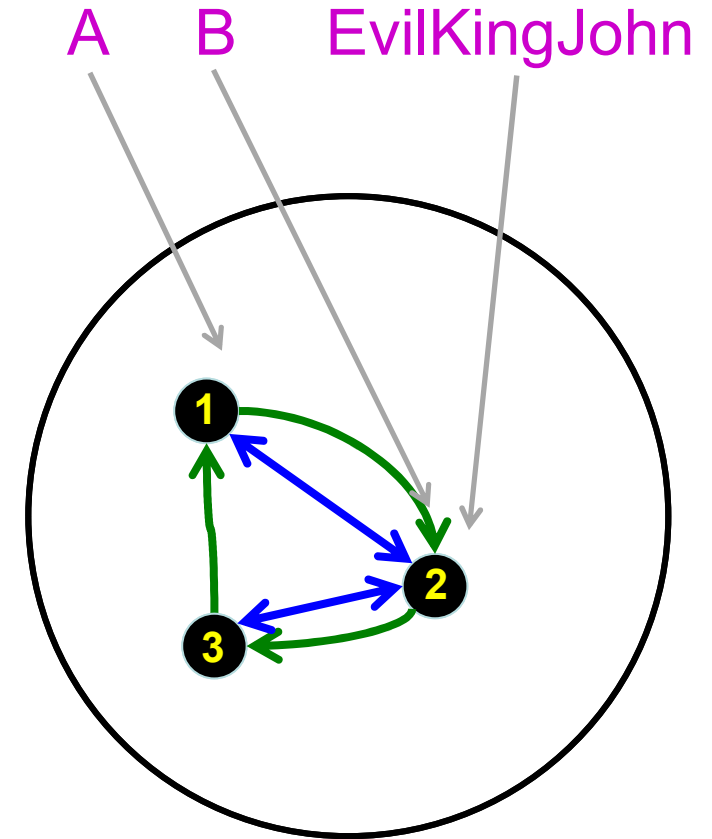
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How many possible worlds?



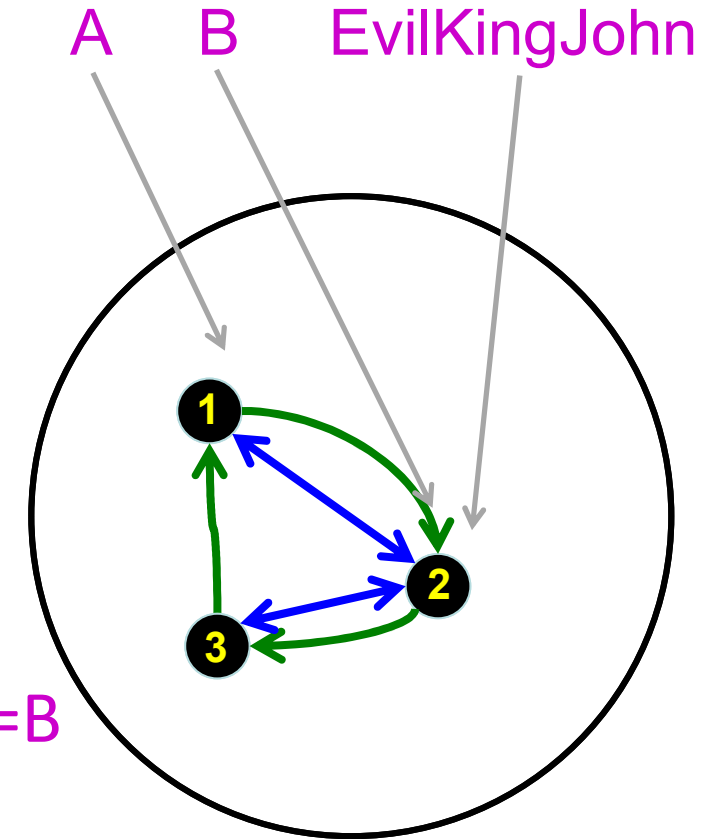
Syntax and semantics: Terms

- A term is something that **refers to an object**; it can be
 - A **constant symbol**, e.g., **A** , **B**, **EvilKingJohn**
 - The possible world fixes these referents
 - A **function symbol** with terms as arguments, e.g., **BFF(EvilKingJohn)**
 - The possible world specifies the value of the function, given the referents of the terms
 - **BFF(EvilKingJohn)** -> **BFF(2)** -> **3**
 - A **logical variable**, e.g., **x**



Syntax and semantics: Atomic sentences

- An atomic sentence is an **elementary proposition** (cf symbols in PL)
 - A **predicate symbol** with terms as arguments, e.g., $\text{Knows}(A, \text{BFF}(B))$
 - $\text{Knows}(A, \text{BFF}(B)) \rightarrow \text{Knows}(1, \text{BFF}(2)) \rightarrow \text{Knows}(1, 3) \rightarrow \mathbf{F}$
 - True iff the objects referred to by the terms are in the relation referred to by the predicate
 - An **equality between terms**, e.g., $\text{BFF}(\text{BFF}(\text{BFF}(B))) = B$
 - True iff the terms refer to the same objects
 - $\text{BFF}(\text{BFF}(\text{BFF}(B))) = B \rightarrow \text{BFF}(\text{BFF}(\text{BFF}(2))) = 2 \rightarrow \text{BFF}(\text{BFF}(3)) = 2 \rightarrow \text{BFF}(1) = 2 \rightarrow 2 = 2 \rightarrow \mathbf{T}$



Syntax and semantics: Complex sentences

1. Sentences with logical connectives

$\neg\alpha, \alpha \wedge \beta, \alpha \vee \beta, \alpha \Rightarrow \beta, \alpha \Leftrightarrow \beta$

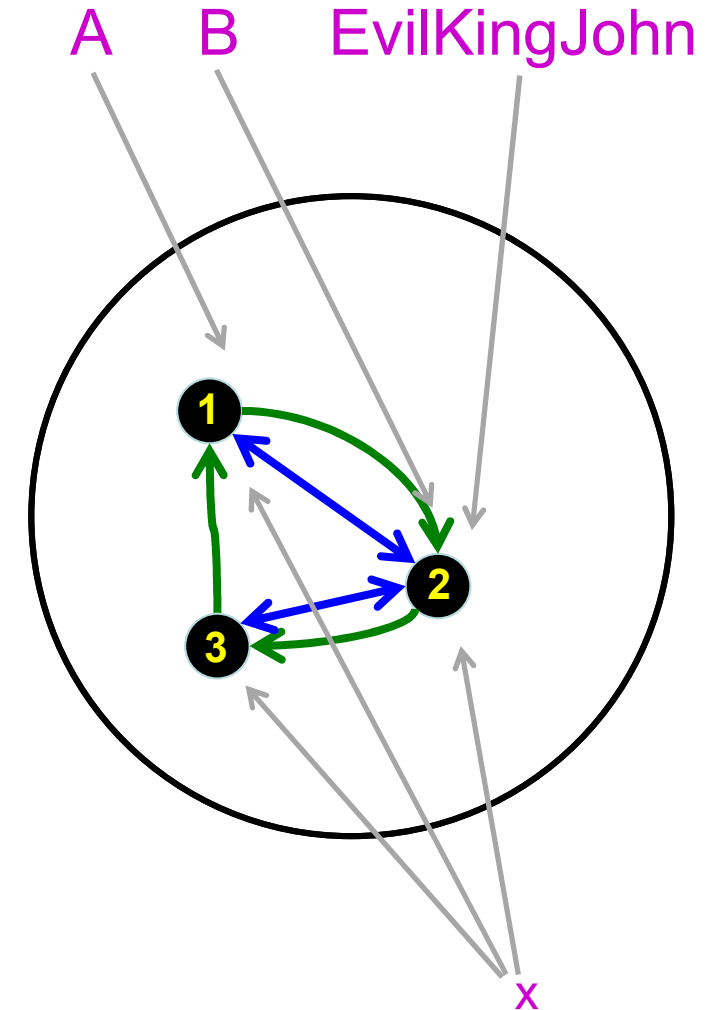
- We want to express properties of entire collections of objects, instead of enumerating the objects by name

2. Sentences with universal or existential quantifiers, e.g.,

- $\forall x \text{ Knows}(x, \text{BFF}(x))$

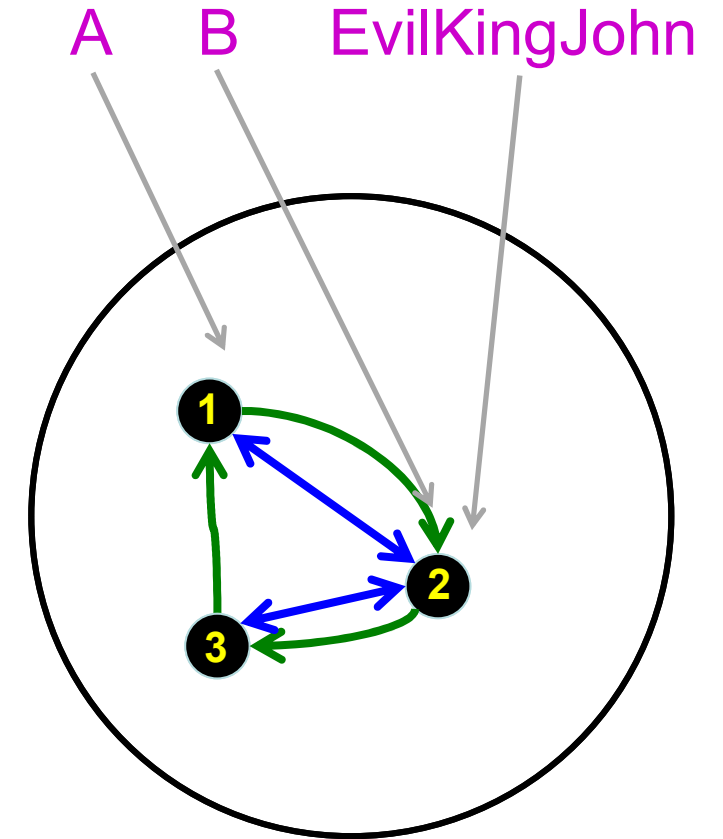
- True in world w iff true in **all extensions** of w where x refers to an object in w

- $x \rightarrow 1: \text{Knows}(1, \text{BFF}(1)) \rightarrow \text{Knows}(1, 2) \rightarrow \text{T}$
- $x \rightarrow 2: \text{Knows}(2, \text{BFF}(2)) \rightarrow \text{Knows}(2, 3) \rightarrow \text{T}$
- $x \rightarrow 3: \text{Knows}(3, \text{BFF}(3)) \rightarrow \text{Knows}(3, 1) \rightarrow \text{F}$



Syntax and semantics: Complex sentences

- Sentences with logical connectives
 $\neg\alpha, \alpha \wedge \beta, \alpha \vee \beta, \alpha \Rightarrow \beta, \alpha \Leftrightarrow \beta$
- Sentences with universal or existential quantifiers, e.g.,
 - $\exists x \text{ Knows}(x, \text{BFF}(x))$
 - True in world w iff true in **some extension** of w where x refers to an object in w
 - $x \rightarrow 1: \text{Knows}(1, \text{BFF}(1)) \rightarrow \text{Knows}(1, 2) \rightarrow \text{T}$
 - $x \rightarrow 2: \text{Knows}(2, \text{BFF}(2)) \rightarrow \text{Knows}(2, 3) \rightarrow \text{T}$
 - $x \rightarrow 3: \text{Knows}(3, \text{BFF}(3)) \rightarrow \text{Knows}(3, 1) \rightarrow \text{F}$



Syntax of FOL

<i>Sentence</i>	$\rightarrow$	<i>AtomicSentence</i> <i>ComplexSentence</i>
<i>AtomicSentence</i>	$\rightarrow$	<i>Predicate</i> <i>Predicate</i> (<i>Term</i> , ...) <i>Term</i> = <i>Term</i>
<i>ComplexSentence</i>	$\rightarrow$	(<i>Sentence</i>) [<i>Sentence</i>] $\neg$ <i>Sentence</i> <i>Sentence</i> $\wedge$ <i>Sentence</i> <i>Sentence</i> $\vee$ <i>Sentence</i> <i>Sentence</i> $\Rightarrow$ <i>Sentence</i> <i>Sentence</i> $\Leftrightarrow$ <i>Sentence</i> <i>Quantifier</i> <i>Variable</i> , ... <i>Sentence</i>
<i>Term</i>	$\rightarrow$	<i>Function</i> (<i>Term</i> , ...) <i>Constant</i> <i>Variable</i>
<i>Quantifier</i>	$\rightarrow$	$\forall$ $\exists$
<i>Constant</i>	$\rightarrow$	<i>A</i> <i>X</i> ₁ <i>John</i> ...
<i>Variable</i>	$\rightarrow$	<i>a</i> <i>x</i> <i>s</i> ...
<i>Predicate</i>	$\rightarrow$	<i>True</i> <i>False</i> <i>After</i> <i>Loves</i> <i>Raining</i> ...
<i>Function</i>	$\rightarrow$	<i>Mother</i> <i>LeftLeg</i> ...

OPERATOR PRECEDENCE : $\neg, =, \wedge, \vee, \Rightarrow, \Leftrightarrow$

Truth in first-order logic

- Sentences are true with respect to a **model** and an **interpretation**
- Model contains objects (**domain elements**) and relations among them
- Interpretation specifies referents for
 - constant symbols $\rightarrow$ objects
 - predicate symbols $\rightarrow$ relations
 - function symbols $\rightarrow$ functional relations
- An atomic sentence $\text{predicate}(\text{term}_1, \dots, \text{term}_n)$ **is true** iff the objects referred to by $\text{term}_1, \dots, \text{term}_n$ are in the relation referred to by predicate

A common mistake to avoid

- Typically, $\Rightarrow$ is the main connective with $\forall$

- “Everyone at KNTU is smart”:

$$\forall x \text{ At}(x, \text{KNTU}) \Rightarrow \text{Smart}(x)$$

- **Common mistake:** using $\wedge$ as the main connective with $\forall$:

$$\forall x \text{ At}(x, \text{KNTU}) \wedge \text{Smart}(x)$$

means “Everyone is at KNTU and everyone is smart”

Another common mistake to avoid

- $\exists$ <variables> <sentence>

- “Someone at KNTU is smart”:

$$\exists x \text{ At}(x, \text{KNTU}) \wedge \text{Smart}(x)$$

- Typically, $\wedge$ is the main connective with $\exists$
- **Common mistake**: using $\Rightarrow$ as the main connective with $\exists$:

$$\exists x \text{ At}(x, \text{KNTU}) \Rightarrow \text{Smart}(x)$$

is true if there is anyone who is not at KNTU!

-
- $\forall x P(x)$ is true in a model m iff $P(x)$ is true with x being each possible object in the model
 - Equivalent to the **conjunction of instantiations** of $P(x)$
 - $At(KingJohn, KNTU) \Rightarrow Smart(KingJohn)$
 $\wedge At(Richard, KNTU) \Rightarrow Smart(Richard)$
 $\wedge At(KNTU, KNTU) \Rightarrow Smart(KNTU)$
 $\wedge \dots$
 - $\exists x P(x)$ is true in a model m iff $P(x)$ is true with x being some possible object in the model
 - Equivalent to the **disjunction of instantiations** of $P(x)$
 - $At(KingJohn, KNTU) \wedge Smart(KingJohn)$
 $\vee At(Richard, KNTU) \wedge Smart(Richard)$
 $\vee At(KNTU, KNTU) \wedge Smart(KNTU)$
 $\vee \dots$

Properties of quantifiers

- $\forall x \forall y \equiv \forall y \forall x \equiv \forall x, y$
- $\exists x \exists y \equiv \exists y \exists x \equiv \exists x, y$
- $\exists x \forall y$ is **not** the same as $\forall y \exists x$
 - $\exists x \forall y \text{ Loves}(x, y)$

“There is a person who loves everyone in the world”
 - $\forall y \exists x \text{ Loves}(x, y)$

“Everyone in the world is loved by at least one person”
- **Quantifier duality**: each can be expressed using the other

$\forall x \text{ Likes}(x, \text{IceCream})$	$\neg \exists x \neg \text{Likes}(x, \text{IceCream})$
$\exists x \text{ Likes}(x, \text{Broccoli})$	$\neg \forall x \neg \text{Likes}(x, \text{Broccoli})$

Properties of quantifiers

- Because $\forall$ is really a conjunction over the universe of objects and $\exists$ is a disjunction, it should not be surprising that they obey De Morgan's rules.
- The De Morgan rules for quantified and unquantified sentences are as follows:

$$\forall x \neg P \equiv \neg \exists x P$$

$$\neg \forall x P \equiv \exists x \neg P$$

$$\forall x P \equiv \neg \exists x \neg P$$

$$\exists x P \equiv \neg \forall x \neg P$$

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$P \wedge Q \equiv \neg(\neg P \vee \neg Q)$$

$$P \vee Q \equiv \neg(\neg P \wedge \neg Q)$$

Fun with sentences

- Everyone knows President Obama
 - $\forall n \text{ Person}(n) \Rightarrow \text{Knows}(n, \text{Obama})$
- Everyone knows someone
 - $\forall x \text{ Person}(x) \Rightarrow \exists y \text{ Person}(y) \wedge \text{Knows}(x, y)$
- There is someone that nobody else knows
 - $\exists s \text{ Person}(s) \wedge \forall n (\text{Person}(n) \wedge \neg(n = s)) \Rightarrow \neg \text{Knows}(n, s)$

More fun with sentences

- Any two people of the same nationality speak a common language
 - $\text{Nationality}(x,n)$ – x has nationality n
 - $\text{Speaks}(x,l)$ – x speaks language l
 - $\forall x,y [(\exists n \text{ Nationality}(x,n) \wedge \text{Nationality}(y,n)) \Rightarrow (\exists l \text{ Speaks}(x,l) \wedge \text{Speaks}(y,l))]$
- $\forall t (\text{Alive}(t) \Leftrightarrow [\text{Alive}(t-1) \wedge \neg \exists g,x,y [\text{Ghost}(g) \wedge \text{At}(\text{Pacman},x,y,t-1) \wedge \text{At}(g,x,y,t-1)])])$

An alternative semantics?

- We believe that Richard has two brothers, John and Geoffrey
- Wrong assertion
 - $\text{Brother}(\text{John}, \text{Richard}) \wedge \text{Brother}(\text{Geoffrey}, \text{Richard})$
 - This assertion is true in a model where Richard has only one brother
 - Add $\text{John} \neq \text{Geoffrey}$
 - The sentence doesn't rule out models in which Richard has many more brothers besides John and Geoffrey
- Correct assertion
 - $\text{Brother}(\text{John}, \text{Richard}) \wedge \text{Brother}(\text{Geoffrey}, \text{Richard}) \wedge \text{John} \neq \text{Geoffrey}$
 $\wedge \forall x \text{ Brother}(x, \text{Richard}) \Rightarrow (x = \text{John} \vee x = \text{Geoffrey})$
- Humans may make mistakes in translating their knowledge into first-order logic, resulting in unintuitive behaviors from logical reasoning systems that use the knowledge.

Conciseness of first order logic

- Pacman can't be in two places at once
 - FOL: $\forall x_1, y_1, x_2, y_2, t (At(x_1, y_1, t) \wedge At(x_2, y_2, t)) \Rightarrow (x_1 = x_2 \wedge y_1 = y_2)$
 - PL: $\neg (At_1,1_0 \wedge At_1,2_0) \wedge \neg (At_1,1_0 \wedge At_1,3_0) \wedge \neg (At_1,1_0 \wedge At_2,1_0) \wedge \neg (At_1,1_0 \wedge At_2,2_0) \wedge \neg (At_1,1_0 \wedge At_2,3_0) \wedge \neg (At_1,1_0 \wedge At_3,1_0) \wedge \neg (At_1,1_0 \wedge At_3,2_0) \wedge \neg (At_1,1_0 \wedge At_3,3_0) \wedge \dots$
 - And that's just if he's in the bottom left at the first timestep

Inference in FOL

- Entailment is defined exactly as for propositional logic:
 - $\alpha \models \beta$ (“ α entails β ”) iff in every world where α is true, β is also true
 - E.g., $\forall x \text{ Knows}(x, \text{Obama})$ entails $\exists y \forall x \text{ Knows}(x, y)$
- In FOL, we can go beyond just answering “yes” or “no”; given an existentially quantified query ($\exists$), return a **substitution** (or **binding**) for the variable(s) such that the resulting sentence is entailed:
 - KB = $\forall x \text{ Knows}(x, \text{Obama})$
 - Query = $\exists y \forall x \text{ Knows}(x, y)$
 - Answer = Yes, $\sigma = \{y/\text{Obama}\}$
 - Notation: $\alpha\sigma$ means applying substitution σ to sentence α
 - E.g., if $\alpha = \forall x \text{ Knows}(x, y)$ and $\sigma = \{y/\text{Obama}\}$, then $\alpha\sigma = \forall x \text{ Knows}(x, \text{Obama})$

Inference in FOL: Propositionalization

- Convert $(KB \wedge \neg\alpha)$ to PL, use a PL SAT solver to check (un)satisfiability
 - Trick: replace variables with ground terms, convert atomic sentences to symbols
 - $\exists x \text{ Knows}(x, \text{Obama})$
 - $\text{Knows}(X_1, \text{Obama})$
 - Knows_X1_Obama
 - $\forall x \text{ Knows}(x, \text{Obama})$ and $\text{Democrat}(\text{Feinstein})$
 - $\text{Knows}(\text{Obama}, \text{Obama})$ and $\text{Knows}(\text{Feinstein}, \text{Obama})$ and $\text{Democrat}(\text{Feinstein})$
 - $\text{Knows_Obama_Obama} \wedge \text{Knows_Feinstein_Obama} \wedge \text{Democrat_Feinstein}$
 - $\forall x \text{ Knows}(\text{Mother}(x), x)$
 - $\text{Knows}(\text{Mother}(\text{Obama}), \text{Obama})$ and $\text{Knows}(\text{Mother}(\text{Mother}(\text{Obama})), \text{Mother}(\text{Obama}))$
 - Real trick: for $k = 1$ to infinity:
 - Get a set of terms: constants, functions of constants, funcs of funcs of constants, ... up to depth k
 - Propositionalize as if those are all the terms that exist
 - If a contradiction is found, halt; otherwise, continue
 - If FOL sentence is unsatisfiable, will find a contradiction for some finite k (Herbrand); if not, may continue for ever; *semidecidable*

Inference in FOL: Lifted inference

- Apply inference rules directly to first-order sentences, e.g.,
 - KB = $\text{Person}(\text{Socrates})$, $\forall x \text{ Person}(x) \Rightarrow \text{Mortal}(x)$
 - conclude $\text{Mortal}(\text{Socrates})$
 - The general rule is a version of Modus Ponens:
 - Given $\alpha \Rightarrow \beta$ and α' , where $\alpha\sigma = \alpha'\sigma$ for some substitution σ , conclude $\beta\sigma$
 - σ is $\{x/\text{Socrates}\}$
 - Given $\forall x \text{ Knows}(x, \text{Obama})$ and $\forall y, z \text{ Knows}(y, z) \Rightarrow \text{Likes}(y, z)$
 - σ is $\{y/x, z/\text{Obama}\}$, conclude $\text{Likes}(x, \text{Obama})$

Generalized Modus Ponens

Classical Modus Ponens (in Propositional Logic):

If you know:

- $P \rightarrow Q$
- P (P is true)

Then you can infer:

- Q

Let's say you have:

- $P_1 \wedge P_2 \wedge \cdots \wedge P_n \rightarrow Q$ (*rule*)
- $P_1' \wedge P_2' \wedge \cdots \wedge P_n'$ (*facts*)

(that can be unified with the left side)

- If you can **unify** each P_i with P_i' using a substitution σ , then you can infer $Q\sigma$

Example

- Rule: $\text{Person}(x) \wedge \text{Owns}(x, y) \wedge \text{Car}(y) \rightarrow \text{Drives}(x, y)$
- Facts: $\text{Person}(\text{John}), \text{Owns}(\text{John}, \text{Car1}), \text{Car}(\text{Car1})$
- Unify Facts with Rule:
 - σ is $\{ x \rightarrow \text{John}, y \rightarrow \text{Car1} \}$
- **apply** this substitution to the **right-hand side** of the rule
 - $\text{Drives}(\text{John}, \text{Car1})$
- New fact
 - $\text{Drives}(\text{John}, \text{Car1})$

Unification

- **Unification** is the process of finding a substitution for variables such that two terms become identical after the substitution

- **Steps:**

- 1. Initialization:**

- Start with an empty substitution set $\sigma = \{\}$.

- 2. Compare the Two Terms:**

- If both terms are **identical**, no change to the substitution is required.
 - If one term is a **variable**:
 - Check if the variable occurs in the other term, if it does it fails
 - If it doesn't occur in the other term, substitute the variable with the other term.
 - If both terms are **functions**:
 - Compare their function names and arities. If they match, proceed to unify their arguments recursively.
 - If one term is a **constant** and the other is a **different constant**:
 - Unification fails because constants must be identical to be unified.

Example

- $\text{Hates}(\text{Father}(x), \text{Brother}(y), z)$ and $\text{Hates}(\text{Father}(\text{John}), \text{Brother}(\text{Bill}), \text{King}(\text{John}))$
 - $\{ x \rightarrow \text{John}, y \rightarrow \text{Bill}, z \rightarrow \text{King}(\text{John}) \}$
- $\text{Hates}(\text{Father}(x), \text{Brother}(y), x)$ and $\text{Hates}(\text{Father}(\text{John}), \text{Brother}(\text{Bill}), \text{King}(x))$
 - Fails

Resolution in FOL

High-level strategy (same as in propositional logic):

- Convert all formulas to CNF
- Repeatedly apply resolution rule

■ But

- What to do with quantifiers?

Resolution

Conversion to CNF

Input: $\forall x (\forall y \text{ Animal}(y) \rightarrow \text{Loves}(x, y)) \rightarrow \exists y \text{ Loves}(y, x)$

Output: $(\text{Animal}(Y(x)) \vee \text{Loves}(Z(x), x)) \wedge (\neg \text{Loves}(x, Y(x)) \vee \text{Loves}(Z(x), x))$

New to first-order logic:

- All variables (e.g., x) have universal quantifiers by default
- Introduce **Skolem functions** (e.g., $Y(x)$) to represent existential quantified variables

Conversion to CNF (part 1)

Anyone who loves all animals is loved by someone.

Input:

$$\forall x (\forall y \text{Animal}(y) \rightarrow \text{Loves}(x, y)) \rightarrow \exists y \text{Loves}(y, x)$$

Eliminate implications (old):

$$\forall x \neg(\forall y \neg \text{Animal}(y) \vee \text{Loves}(x, y)) \vee \exists y \text{Loves}(y, x)$$

Push $\neg$ inwards, eliminate double negation (old):

$$\forall x (\exists y \text{Animal}(y) \wedge \neg \text{Loves}(x, y)) \vee \exists y \text{Loves}(y, x)$$

Standardize variables (**new**):

$$\forall x (\exists y \text{Animal}(y) \wedge \neg \text{Loves}(x, y)) \vee \exists z \text{Loves}(z, x)$$

Conversion to CNF (part 2)

$$\forall x (\exists y \text{Animal}(y) \wedge \neg \text{Loves}(x, y)) \vee \exists z \text{Loves}(z, x)$$

Replace existentially quantified variables with Skolem functions (**new**):

$$\forall x [\text{Animal}(Y(x)) \wedge \neg \text{Loves}(x, Y(x))] \vee \text{Loves}(Z(x), x)$$

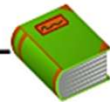
Distribute $\vee$ over $\wedge$ (old):

$$\forall x [\text{Animal}(Y(x)) \vee \text{Loves}(Z(x), x)] \wedge [\neg \text{Loves}(x, Y(x)) \vee \text{Loves}(Z(x), x)]$$

Remove universal quantifiers (**new**):

$$[\text{Animal}(Y(x)) \vee \text{Loves}(Z(x), x)] \wedge [\neg \text{Loves}(x, Y(x)) \vee \text{Loves}(Z(x), x)]$$

After converting all formulas to CNF, then we can apply the resolution rule, which is generalized to first-order logic. This means that instead of doing exact matching of a literal p , we unify atomic formulas p and q , and then apply the resulting substitution θ on the conclusion.



Definition: resolution rule (first-order logic)

$$\frac{f_1 \vee \dots \vee f_n \vee \textcolor{red}{p}, \quad \neg \textcolor{red}{q} \vee g_1 \vee \dots \vee g_m}{\text{Subst}[\theta, f_1 \vee \dots \vee f_n \vee g_1 \vee \dots \vee g_m]}$$

where $\theta = \text{Unify}[p, q]$.



Example: resolution

$$\frac{\text{Animal}(Y(x)) \vee \text{Loves}(Z(x), x), \quad \neg \text{Loves}(u, v) \vee \text{Feeds}(u, v)}{\text{Animal}(Y(x)) \vee \text{Feeds}(Z(x), x)}$$

Substitution: $\theta = \{u/Z(x), v/x\}$.

Quiz questions

- “Translate the following sentences into logic, using intuitively named functions and relations/predicates.
 - E.g. Every person knows Obama : $\forall n \text{ Person}(n) \Rightarrow \text{Knows}(n, \text{Obama})$
- Sentences:
 - Someone is called Steve
 - $\exists x \text{ CalledSteve}(x)$ or $\exists x \text{ Person}(x) \wedge \text{Name}(x) = \text{Steve}$
 - Everybody hates everybody else
 - $\forall x, y \neg(x = y) \Rightarrow \text{Hates}(x, y)$
 - If someone has two brothers, they’re brothers with each other
 - $\forall x, y, z (\text{Brother}(x, y) \wedge \text{Brother}(x, z) \Rightarrow \text{Brother}(y, z))$

Quiz Questions

- A possible world of five objects:

 "left leg" unary function (arity is # arguments)

 "on head" binary relation

 "brother" binary relation

 "person" unary relation

 "king" unary relation

 "crown" unary relation

 "John" constant (0-ary function)

 "Richard" constant (0-ary function)

- Which of following sentences are true in this possible world?
- Which one is not a sentence (syntactically incorrect)?
- $\exists x, y \text{ Brother}(x, \text{Richard}) \wedge \text{OnHead}(y, x)$
- $\neg (\text{LeftLeg}(\text{Richard}) = \text{LeftLeg}(\text{John}))$
- $\text{Brother}(\text{John}) = \text{Richard}$
- $\forall x, y (\text{Person}(x) \wedge \text{Person}(y)) \Rightarrow \text{Brother}(x, y)$
- $\forall x \text{ Person}(x) \wedge (\exists y \text{ Crown}(y) \wedge \text{OnHead}(y, x)) \Leftrightarrow \text{King}(x)$

